

# Ethan W. Todd

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## Education

### University of Michigan

*Candidate, Ph.D. Physics*

Advisors: Prof. Igor Jovanovic, Prof. James Wells

2025 M.S. Physics  
Winter

Ann Arbor, MI

2023 – Present (expected 2026)

Fall

Fall

### Florida State University

*B.S. Physics, Summa Cum Laude*

Honors Thesis: Study of Displaced Vertex Tagging with the CMS Experiment

Advisor: Prof. Ted Kolberg

Tallahassee, FL

2020 – 2023

Fall

Spring

### Emory University

*Foundational Coursework*

Atlanta, GA

2018 – 2020

Fall

Spring

## Publications

6. V. Fondement, J. Bae, F. Rahman, et al., “Digital Pulse Processing for Mixed-Field Ionizing Radiation Detection in Rare-Event Nuclear Experiments,” *under review*, 2026
5. V. Fondement, **E. Todd**, and I. Jovanovic, “Measurement of  $^{78m}\text{Br}$   $2^- \rightarrow 1^+$  Transition Half-Life by Self-Coincidence Analysis in an Active Target  $\text{LaBr}_3(\text{Ce})$  Detector,” *under review*, 2026
4. **E. Todd**, V. Fondement, and I. Jovanovic, “Machine Learning-Enabled Pulse Shape Discrimination in  $\text{YAP:Ce}$ ,” *Nuclear Instruments and Methods in Physics Research Section A*, 2026, accepted
3. CMS HGCAL Collaboration, “Leakage Current of High-Fluence Neutron-Irradiated 8” Silicon Sensors for the CMS Endcap Calorimeter Upgrade,” *under review*, 2026. [Online]. Available: <https://arxiv.org/abs/2510.17280>
2. C. Graham, **E. Todd**, C. Wilson, et al., “Light Output-Dependent Integration Gating for Optimized Pulse Shape Discrimination,” *Nuclear Instruments and Methods in Physics Research Section A*, 2026. DOI: [10.1016/j.nima.2026.171521](https://doi.org/10.1016/j.nima.2026.171521)
1. K. Yocum, H. Smith, **E. Todd**, et al., “Millimeter/Submillimeter Spectroscopic Detection of Desorbed Ices: A New Technique in Laboratory Astrochemistry,” *The Journal of Physical Chemistry A*, 2019. DOI: [10.1021/acs.jpca.9b04587](https://doi.org/10.1021/acs.jpca.9b04587)

## Research Experience

### Low Energy Physics Group, Argonne National Laboratory

*Graduate Research Aide*

Lemont, IL

2026 – Present

Summer

### Measurement of the $^{37}\text{Cl}(\alpha, n)^{40}\text{K}$ Cross Section

Supervisor: Dr. Daniel Santiago-Gonzalez

- Performed experimental measurements of the  $^{37}\text{Cl}(\alpha, n)^{40}\text{K}$  reaction using the Multi-Sampling Ionization Chamber (MUSIC) at the ATLAS accelerator facility at Argonne National Laboratory.
- Developed an event reconstruction algorithm for prior MUSIC data that performs offline time synchronization of four digitizers via analysis of the time structure of the  $^{37}\text{Cl}$  beam, and uses CUDA-accelerated time sorting to greatly increase data-processing throughput.
- Currently searching for  $^{37}\text{Cl}(\alpha, n)^{40}\text{K}$  reaction events in a newly acquired MUSIC dataset while investigating methods to improve the quality of the prior dataset, including by performing unsupervised classification with ML clustering algorithms.

**Applied Nuclear Science Group, University of Michigan**  
*Graduate Student Researcher*

**Ann Arbor, MI**  
**2024 – Present**  
*Summer*

### Particle Identification in YAP:Ce via Machine Learning

- Compared  $\alpha$  and  $\gamma$ -ray discrimination capabilities of charge comparison/shape indicator to several machine learning methods: random forests, gradient boosting, XGBoost, and multilayer perceptrons.
- Quantified the performance of each method using receiver operating characteristic curves, and demonstrated the substantial advantage of machine learning over traditional methods.
- Applied the XGBoost model to proof-of-concept mixed field radiation environments consisting of  $^{241}\text{Am}$  and  $^{137}\text{Cs}$  or  $^{241}\text{Am}$  and  $^{60}\text{Co}$ , and demonstrated that known spectral features of  $^{241}\text{Am}$ ,  $^{137}\text{Cs}$ , and  $^{60}\text{Co}$  are correctly classified.

### Precision Measurements of $^{72}\text{Ge}(n, \gamma)$ at Thermal Energies

- Utilized the Ohio State Nuclear Reactor Laboratory thermal neutron beam facility to perform new measurements of  $^{72}\text{Ge}(n, \gamma)^{73}\text{Ge}$  gamma-ray transition energies with a focus on the 68.75 keV final-state transition.
- Used a **position-sensitive H3D M400 CdZnTe detector** to measure low-energy gamma-ray interactions with event-by-event position sensitivity.
- Developed custom analysis code to select single-scatter events near the detector surface, reducing the Compton background and improving sensitivity in the low-energy region.
- Provided an independent measurement of the 68.75 keV gamma-ray energy in order to verify a separate UZH/PSI gamma-ray measurement and further support the validity of a recent high 254 eV<sub>nr</sub> germanium ionization-yield result.

**CMS Group, Florida State University**  
*Undergraduate Student Researcher*

**Tallahassee, FL**  
**2021 – 2023**  
*Fall Summer*

### Search for BSM Long-lived Particles

Advisor: Prof. Ted Kolberg

- Utilized deep neural network machine learning to search for beyond the Standard Model long-lived particles in data from the CMS Experiment.
- Trained ML models on Monte Carlo data with varying signal (mass, lifetime) and background ( $t, b$  physics) to verify their ability to identify displaced vertices independently of the particular physics.
- Investigated the most effective parameters of ML models and verified physics-based expectations.

### Silicon Sensor Quality Control

Advisor: Prof. Rachel Yohay

- Conducted process quality control (PQC) testing for the CMS High Granularity Calorimeter (HGCAL) to monitor the quality and stability of the silicon sensor production process.
- Analyzed and presented PQC results to the CMS HGCAL sensor testing group.
- Helped create and implement a standardized testing procedure at FSU.

**WW Astrochemistry Group, Emory University**  
*Undergraduate Student Researcher*

**Atlanta, GA**  
**2018 – 2020**  
*Fall Fall*

Advisor: Prof. Susanna Widicus Weaver

- Performed millimeter/submillimeter spectroscopic measurements of desorbed astrophysical ice analogs.
- Conducted proof-of-concept millimeter/submillimeter spectroscopic measurements of  $\text{H}_2\text{O}$  and  $\text{D}_2\text{O}$  binding energy, sublimation enthalpy, and sublimation entropy.

## Teaching Experience

**University of Michigan**  
*Lead Graduate Student Instructor*

**Ann Arbor, MI**  
**2025 – 2026**  
*Fall Winter*

- Oversee 30 graduate student instructors teaching general physics laboratories with a total enrollment of 1800 undergraduate students.
- Edit and maintain the laboratory manuals and Jupyter notebook worksheets used in the course.
- Manage the course information and organization via Canvas.
- Arbitrate student and graduate student instructor concerns.

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**University of Michigan**  
*Graduate Student Instructor*

**Ann Arbor, MI**  
2023–2025  
*Fall Spring*

- 2025 Spring Physics 251, Fundamental Physics for the Life Sciences II Lab
- 2024 Fall Physics 121, Physics for Architects Lab
- 2023 – 2024 Fall Winter Physics 241, General Physics II Lab

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**Florida State University**  
*Physics Learning Assistant*

**Tallahassee, FL**  
2022  
*Spring*

PHY 2049C, General Physics B

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**Emory University**  
*Laboratory Teaching Assistant*

**Atlanta, GA**  
2019  
*Fall*

CHEM 150L, Structure and Properties Lab

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## Honors and Awards

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|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>2026</b> <small>Winter</small> : Marcellus L. Wiedenbeck Teaching Award                | <i>Awarded for exemplifying outstanding teaching abilities.</i>                             |
| <b>2026</b> <small>Winter</small> : Rackham Outstanding Graduate Student Instructor Award | <i>Awarded for demonstrating extraordinary dedication and excellence as a teacher.</i>      |
| <b>2023</b> <small>Spring</small> : Phi Beta Kappa Society                                |                                                                                             |
| <b>2023</b> <small>Spring</small> : Joseph Lanutti Undergraduate Research Award           | <i>Awarded for third place in the FSU physics departmental poster presentation contest.</i> |
| <b>2022</b> <small>Fall</small> : Evelyn and John Baugh Research Presentation Scholarship | <i>Awarded to travel and present research at SESAPS.</i>                                    |
| <b>2022</b> <small>Spring</small> : Anna Runyan Undergraduate Endowment                   | <i>Awarded for outstanding academic success in physics.</i>                                 |

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## Grants

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| <b>2025</b> <small>Fall</small> : NSUF Rapid Turnaround Experiment Grant  | <i>Awarded to conduct precision measurements of Ge gamma-ray cascades at the Ohio State University Nuclear Reactor Laboratory.</i> |
| <b>2025</b> <small>Fall</small> : Rackham Graduate Student Research Grant | <i>Awarded to conduct research at the Paul Scherrer Institute on the ionization yield of high-purity germanium detectors.</i>      |

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## Skills

**Programming:** C++, Python, Nix, CUDA

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|------------------|--------------------------|-------------------------------------------------------------------------|
| <b>Software:</b> | Nuclear Measurements:    | ROOT, Geant4, CoMPASS, wavedump, <b>LISE<sup>++</sup></b> , <b>SRIM</b> |
|                  | Machine Learning:        | Scikit-learn, XGBoost, TensorFlow                                       |
|                  | Artificial Intelligence: | Local deployment of open-source LLMs as agents via <b>llama.cpp</b>     |
|                  | Version Control:         | Git, <b>Nix Flakes</b>                                                  |
|                  | Build Systems:           | <b>Nix</b> , Make, CMake                                                |
|                  | Scientific Presentation: | <b>L<sup>A</sup>T<sub>E</sub>X</b> /Beamer                              |

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**Hardware:** CAEN digitizers, soldering, electronic testing equipment, *e.g.*, oscilloscopes

## Presentations

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5. **E. Todd**, V. Fondement, and I. Jovanovic, *Machine Learning-Enabled Pulse Shape Discrimination in Fast Inorganic Scintillators*, ANL PSE Student Lunch Talk Summer Series, Lemont, IL, 2026
4. **E. Todd**, V. Fondement, and I. Jovanovic, *Particle Identification in Fast Inorganic Scintillators via Machine Learning*, Fall Meeting of the Eastern Great Lakes Section of the American Physical Society, Ypsilanti, MI, 2025
3. **E. Todd** and I. Jovanovic, *Investigation of Anomalously High Ionization Produced by Low Energy Nuclear Recoils in Germanium (poster)*, DOE NNSA Office of Defense Nuclear Nonproliferation R&D University Program Review, Gainesville, FL, 2025
2. **E. Todd**, A. A. Kadhim, N. Bower, et al., *Process Quality Control for HGCal at Florida State University (poster)*, 89th Annual Meeting of the Southeastern Section of the American Physical Society, Oxford, MS, 2022
1. **E. Todd**, K. Yocum, P. Gerakines, et al., *A Novel Use of Rotational Spectroscopy for Studying Characteristics of Desorbed Interstellar Ice Analogs (poster)*, Emory SURE 2019 Symposium, Atlanta, GA, 2019